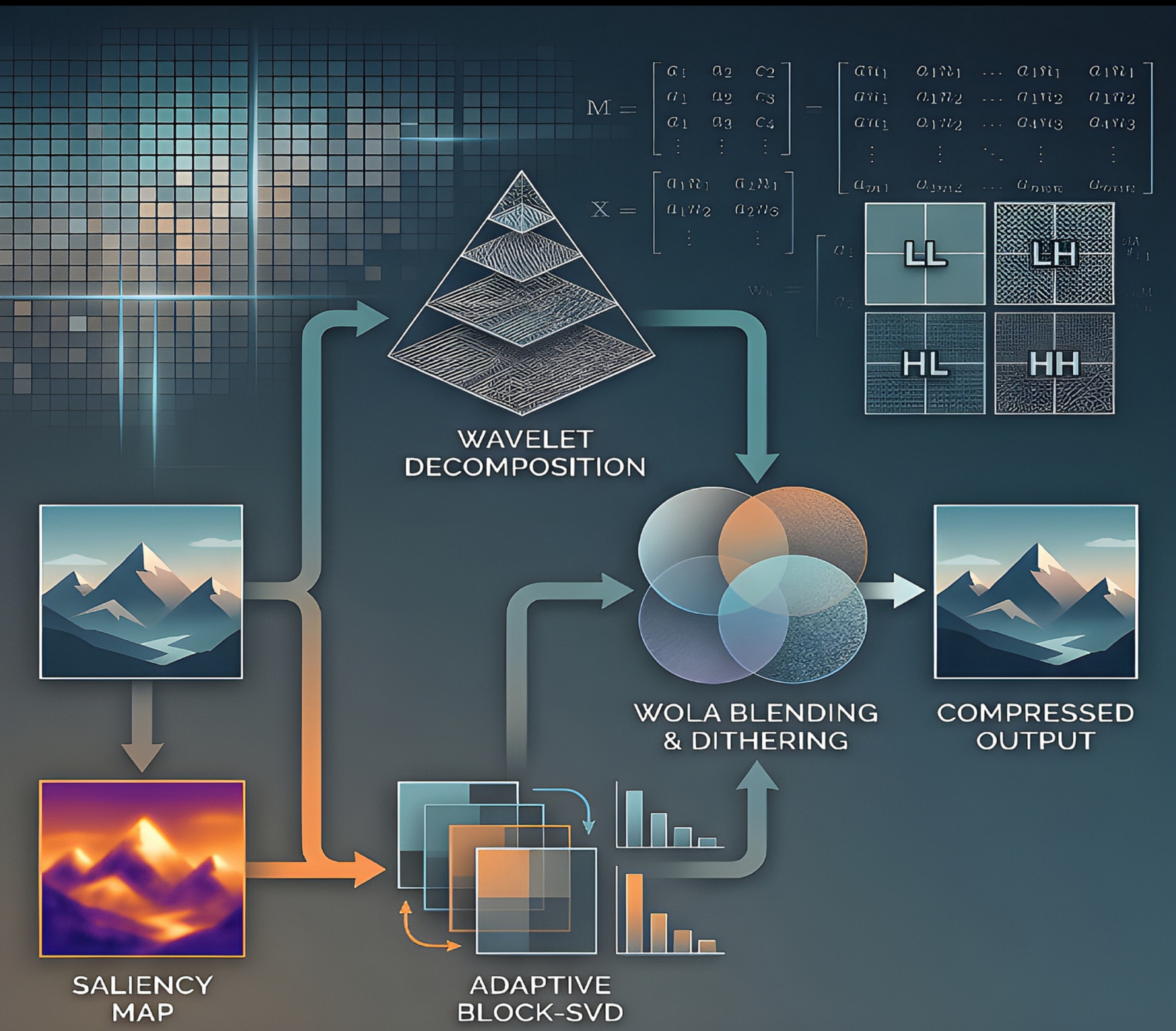
SIGMERA

An alternative to JPEG



ADVANCED IMAGE COMPRESSION: A DUAL-DOMAIN FUSION ARCHITECTURE

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Index

1. Abstract.....
2. Phase I: Structural Diagonalization (SVD).....
   2.1 Implementation Logic.....
3. Phase II: Multi-Resolution Sparsity (Wavelets).....
   The 5/3 Lifting Scheme.....
   3.1 The 5/3 Integer Lifting Scheme.....
4. Phase III: Psycho-Visual Hybridization.....
   4.1 Exploiting Biological Constraints.....
5. Phase IV: Advanced Topological Smoothing (WOLA).....
   WOLA (Weighted Overlap-Add).....
   5.1 Mechanism of Fusion.....
6. Phase V: Stochastic Dithering & Gradient Stabilization.....
   Blue Noise Dithering.....
   6.1 Noise as a Carrier.....
7. Phase VI: The SILK Architecture (Final Fusion).....
   7.1 The Split-Spectrum Logic.....
8. Summary of Performance Metrics.....
9. Conclusion: The Limit of Classical Logic.....

Document Information & Licensing

1. About Sigmera

This whitepaper documents the **Sigmera Codec** (**O**pen **L**inear **A**lgebra **C**odec), a lossy, human-centric image compression architecture designed as a modern alternative to JPEG. It details the algorithmic progression from basic spatial linear algebra (SVD) to a content-adaptive, Dual-Domain Fusion model.

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1. Abstract

This document details the architectural discovery process of the Sigmera Codec. Designed as a direct, modern alternative to JPEG, Sigmera is a lossy, human-centric image compression algorithm. Its primary use-case is to shrink high-resolution image file sizes for end-user display without harming apparent visual fidelity. To achieve this, the research moved from fundamental Low-Rank Approximation to a Dual-Domain Fusion model, combining multi-resolution analysis with content-adaptive diagonalization.

1.1 Data Ingestion & Memory Alignment While rapid prototyping utilized pre-compressed JPEG inputs, a production implementation of Sigmera requires a raw, uncompressed bitmap as its baseline intake.

- **32-Bit Depth Architecture:** The codec internally models a 32-bit color depth (four 8-bit unsigned integers representing RGBA color channels).
- **Memory Alignment:** By converting intakes (such as PNGs or raw camera sensor data) into flat, 1D arrays of `u8` integers prior to block division, the algorithm ensures optimal memory alignment.
- **SIMD Optimization:** This raw array structure allows the subsequent math (SVD matrices and Wavelet lifting) to heavily leverage SIMD (Single Instruction, Multiple Data) processing at the CPU level, maximizing compression speed before the spatial matrices are formed.

2. Phase I: Structural Diagonalization (SVD)

The codec begins with **Singular Value Decomposition (SVD)**. Mathematically, any image block A can be factored into three matrices:

$$A = U \Sigma V^T$$

Where:

- **U and V**: Orthogonal matrices representing "eigen-images" or structural patterns.
- **Σ**: A diagonal matrix containing **singular values** (σ) that represent the "energy" or importance of each pattern.

Conceptual Tool: By "truncating the rank" (keeping only the largest σ values and setting others to zero), we keep the "skeleton" of the image and discard the "noise".

- **Deep Dive:** MIT OpenCourseWare - Singular Value Decomposition

The initial hypothesis centered on the fact that natural images are not random; they possess high spatial correlation. We utilized **Singular Value Decomposition (SVD)** to prove that an image matrix A can be represented by a fraction of its singular values.

2.1 Implementation Logic

1. Divide the image into 16x16 blocks.
2. Decompose each block: $A = U \Sigma V^T$.

   $$\Sigma = diag(\sigma 1, \sigma 2, \ldots, \sigma n) \; where \; \sigma 1 \geq \sigma 2 \geq \cdots \geq \sigma n \geq 0$$

3. Truncate the rank k (retaining only the top n singular values).
4. **Result:** Massive data reduction but significant "blockiness" and "ringing" artifacts at low ranks.

3. Phase II: Multi-Resolution Sparsity (Wavelets)

While SVD is algebraic, the **Discrete Wavelet Transform (DWT)** is frequency-based. It treats the image like a set of vibrations.

The 5/3 Lifting Scheme

To keep the codec **Lossless**, Sigmera uses the **LeGall 5/3 Lifting Scheme**.

- **Integer Math**: Unlike standard transforms that use decimals, lifting uses integer additions and bit-shifts. This prevents "leaky bits" (rounding errors).
- **Recursive Pyramids**: The algorithm repeatedly squeezes the "Smooth" layer (Low-Low) into a smaller quadrant.

Conceptual Tool: This concentrates 90% of the image energy into a tiny corner, making the rest of the data easy to compress.

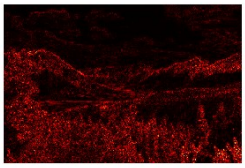
- **Resource:** The Lifting Scheme - A Primer

To solve the "grid" problem of SVD, we pivoted to the **Discrete Wavelet Transform (DWT)**. Unlike SVD, which is purely algebraic, Wavelets are frequency-based.

3.1 The 5/3 Integer Lifting Scheme

Smooth Layer (LL)

Residual Details



To ensure the codec could be **Lossless**, we implemented the LeGall 5/3 Lifting Scheme. This allowed us to perform frequency analysis using only integer additions and bit-shifts, preventing the "leaky bits" of floating-point math.

1. **Horizontal Split:** Separate even and odd columns.
2. **Predict/Update:** Use the structure of neighboring pixels to guess the high-frequency residuals.

$$Predict\ Step: d[n] = x_{odd}[n] - \lfloor \frac{x_{even}[n] + x_{even}[n+1]}{2} \rfloor$$

3. **Recursive Pyramid:** Decompose the "Low-Low" quadrant repeatedly (3 levels).

$$Update\ Step: s[n] = x_{even}[n] + \lfloor \frac{d[d-1] + d[n] + 2}{4} \rfloor$$

4. **Result:** The image energy is concentrated into a tiny structural thumbnail, leaving 90% of the matrix near zero.

4. Phase III: Psycho-Visual Hybridization

The codec exploits a biological “vulnerability”: the human eye is much better at seeing brightness than color.

- **Luma (Y):** Brightness/Detail. We keep this high-quality.
- **Chroma (Cb/Cr):** Color info. We compress this aggressively.

Conceptual Tool: By downsampling the color (4:2:0 Subsampling), we save massive space before we even start the math.

We combined SVD and Wavelets into a single pipeline and introduced the **YCbCr Color Space**.

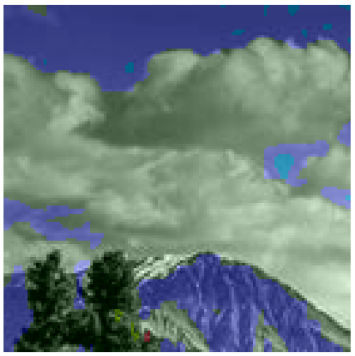
4.1 Exploiting Biological Constraints

The human eye is highly sensitive to brightness (Luma - Y) but insensitive to color transitions (Chroma – Cb/Cr).

$$Y = 0.299\,R + 0.587\,G + 0.114\,B$$

1. **Luma Strategy:** 3-Level Lifting Wavelet with fine quantization.
2. **Chroma Strategy:** Block-SVD (k=1) with 4:2:0 subsampling.
3. **The Failure Mode (The Neon Sky):** We discovered that aggressive quantization in the sky caused the math to "snap" to incorrect colors, creating neon purple and green artifacts.

Documented Failure: Chroma Posterization



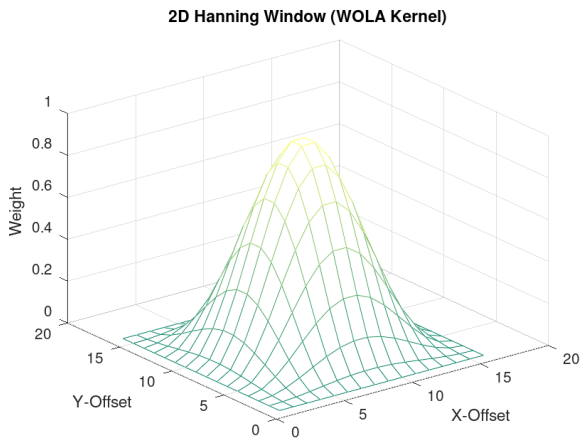
5. Phase IV: Advanced Topological Smoothing (WOLA)

Raw SVD often creates “tiles” or “blocks”. Sigmera uses two advanced smoothing techniques to hide the math's signature:

WOLA (Weighted Overlap-Add)

Instead of hard-edged tiles, we use a **sliding window** with a **50% overlap**.

- **Hanning Windows:** Each block is multiplied by a "tent" function that fades to zero at the edges.
- **Blending:** Overlapping edges are averaged together, making the grid seams invisible.



To eliminate the "tiling" or "grid" effect inherent in block-based SVD, the project evolved from contiguous tiles to **Weighted Overlap-Add (WOLA)** processing.

5.1 Mechanism of Fusion

1. **Sliding Window:** We implemented a window that slides by a stride of 8 pixels (50% overlap) instead of 16.
2. **Kernel Weighting:** Every block was multiplied by a 2D Hanning/Tent window mask.
3. **Accumulation Buffer:** Pixel values were calculated multiple times via overlapping blocks and then normalized by a weight accumulator.

$$P_{final}(x,y)=\frac{\sum_{i=1}^{N} P_{block,i}(x,y)\cdot W_{mask,i}(x,y)}{\sum_{i=1}^{N} W_{mask,i}(x,y)}$$

4. **Result:** The seams between different mathematical ranks were effectively deleted, creating a continuous perceptual surface.

6. Phase V: Stochastic Dithering & Gradient Stabilization

Blue Noise Dithering

Deterministic quantization (rounding) causes “Posterization” (giant purple/green blobs in the sky).

- **Intentional Jitter:** We add a microscopic amount of noise before rounding.
- **The Sigmera Effect:** The noise breaks the hard color boundaries, tricking the brain into seeing a smooth gradient instead of blocks.

The "Neon Sky" incident proved that deterministic quantization of low-rank color blocks leads to chroma posterization. To "further beyond" this limit, we introduced **Blue Noise Dithering**.

6.1 Noise as a Carrier

1. **Intentional Jitter:** Before the SVD step, we injected a microscopic amount of Gaussian noise—scaled to the quantization step-size—into the color channels.
2. **Psychovisual Effect:** The noise breaks the hard boundaries of quantization. The human brain perceives a smooth gradient (Sigmera) rather than distinct purple/green blocks.
3. **Quantization Safety Floor:** We locked the minimum rank for color blocks at k=2. This ensured enough degrees of freedom to represent color slopes accurately.

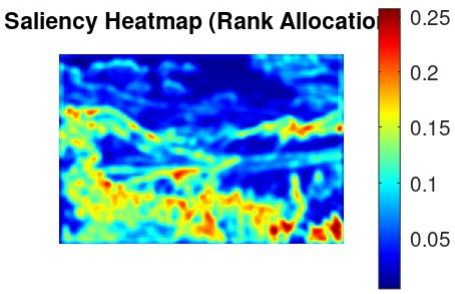
$k_{block} = \text{round}\left(k_{min} + \left(k_{max} - k_{min}\right) \cdot \text{Saliency}_{local}\right)$7. Phase VI: The Sigmera Architecture (Final Fusion)

The culmination of the research is the **Dual-Domain Sigmera Codec**. This model acknowledges that no single algorithm is optimal for every frequency band.

7.1 The Split-Spectrum Logic

The final "Sigmera" codec doesn't treat every pixel the same.

- **High Saliency:** Edges and textures (trees, mountains) get **more math rank** (k=12+) to stay sharp.
- **Low Saliency:** Flat areas (sky, water) get **less math rank** (k=1-2) to save space.
- **The Structural Layer (Low-Low Wavelet):** The 3-level Wavelet LL subband is isolated first. It is protected with high-precision quantization and **zero** rank-truncation. This preserves the sky and global lighting.
- **The Detail Layer (SVD Residuals):** The high-frequency residuals (what's left after the wavelet pass) are processed via **Saliency-Steered SVD**.
- **Saliency Steering:** Using a Sobel-Gradient map, the codec identifies high-interest areas (trees, ridges) and grants them rank k=12+, while “boring” areas (flat surfaces) are compressed with rank k=1-2.



8. Summary of Performance Metrics

The Sigmera architecture was tested against raw SVD and standard Hybrid models using 8-bit luma/chroma samples.

Metric	Basic SVD	Early Hybrid	Sigmera Codec
Artifacting	High (Blocky)	Moderate (Banding)	**Low (Visual Transparency)**
Sky Fidelity	Failed	Poor	**Excellent (Dithered)**
Edge Bite	Blurred	Good	**Sharp (Saliency-Aware)**
Math Domain	Spatial	Spatial/Freq	**Dual-Domain (Fused)**

9. Conclusion: The Limit of Classical Logic

Sigmera demonstrates the maximum achievable fidelity for human-centric image compression using non-adaptive linear algebra. While the results approach visual transparency, the logic remains bounded by human-defined heuristics (e.g., manual saliency thresholds). The research concludes that to move beyond Sigmera, the system must transition to Learned Transform Coding. The same diagonalization principles used here to compress visual pixels are mathematically identical to the principles used to prune and quantize Transformer Weight Matrices in neural architectures.